

PARTS LIST

Semiconductors:

IC1	- TL431/LM431 shunt regulator
IC2	- 7406 hex inverter
IC3	- 7812 voltage regulator
IC4	- 7805 voltage regulator
BR1	- 2A bridge rectifier
T1	- BD139 npn transistor
T2	- TIP31 npn power transistor
D1-D9	- 1N4007 rectifier diode
LED1-LED3	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):	
R1, R4	- 1-kilo-ohm
R2	- 150-ohm, 0.5W
R3	- 3.3-ohm, 1W
R5	- 10-ohm, 2W
R6	- 3.3-ohm, 2W
R7	- 1-ohm, 2W
R8	- 0.33-ohm, 3W
R9, R11, R14	- 2.2-kilo-ohm
R10	- 3.3-kilo-ohm
R12	- 10-kilo-ohm
R13	- 2-kilo-ohm
R15	- 100-ohm, 1W
VR1-VR6	- 47-kilo-ohm potentiometer

Capacitors:

C1-C5, C8, C10-C12, C15, C16	- 0.1µF, ceramic disk
C6	- 2200µF, 35V electrolytic
C7	- 47µF, 35V electrolytic
C9	- 100µF, 35V electrolytic
C13	- 47µF, 25V electrolytic
C14	- 0.1µF, 630V polyester
C17	- 47µF, 16V electrolytic

Miscellaneous:

S1-S4	- On/off switch
F1	- 300mA fuse with holder
CON1	- 2-pin terminal connector
CON2	- 3-pin connector
CON3-CON4	- 2-pin connector
CON5	- 8-pin burg strip male connector
X1	- 230V AC primary to 18V, 2A secondary transformer
	- Heat-sinks for 7812, 7805
	- Heat-sinks for T1 and T2
	- 2-pin connector terminal for X1

or trimmer potentiometers to the required values.

This article presents a programmable power supply built around TL431 (IC1) and two bipolar transistors BD139 and TIP31 (T1 and T2).

The circuit also includes an inverter 7406 (IC2), nine diodes 1N4007 (D1 through D9), a 12V regulator 7812 (IC3), a 5V regulator 7805 (IC4) and a few other components.

TABLE I Digital Codes

Digital codes at CON5 from MCU or from printer port of PC (C0 through C5)	Active potentiometer (VR1 through VR6)
000000	None*
000001	VR1
000010	VR2
000100	VR3
001000	VR4
010000	VR5
100000	VR6

*Note: Vout is around 2.5V in this case and can be set with one more trimmer potentiometer (not shown here)

TABLE II Current-Limiting Functions Using Switches

Closed switch	Output current
None	Around 60mA to 80mA
S1	Around 200mA to 250mA
S2	Around 600mA to 800mA
S3	Around 1A or more

TABLE III Test Points

Test point	Details
TP0	0V (GND)
TP1	Around 18V-20V
TP2	3V-15V
TP3	12V
TP4	5V

Using this circuit you can obtain around 18V, 2A unregulated output and 3V to 15V, 1A variable regulated power supply based on digitally programmable input as shown in Table I. You can also obtain around 12V and 5V, 1A regulated output across connectors CON3 and CON4, respectively.

Circuit and working

Circuit diagram of the programmable power supply is shown in Fig. 1. The mains power supply is applied to transformer X1, which is stepped

down to 18V AC, 2A and is given to bridge rectifier BR1. Fuse F1, resistor R15 and capacitor C14 protect X1 from external surge voltage. C6 is the main filtering capacitor, which should be at least 2200µF.

Voltage produced by TL431 is adjusted with potentiometers VRx (VR1 through VR6) and can be calculated with the simplified formula given below:

$$V_{out} = 2.5 \times (1 + R12 / (R13 + VRx))$$

The maximum output voltage (Voutmax) is thus calculated as:

$$V_{outmax} = 2.5 \times (1 + 10k/2k) = 2.5 \times 6 = 15V$$

Because in this case we have R12 = 10-kilo-ohm and R13 = 2-kilo-ohm. The minimum output voltage (Voutmin) is calculated as:

$$V_{outmin} = 2.5 \times (1 + 10k / (2k + 47k)) = 2.5 \times 1.2 = 3V$$

Because in this case we have R12 = 10-kilo-ohm, R13 = 2-kilo-ohm and VRx = 47-kilo-ohm.

Table I shows how output voltages can be set with digital codes C0 through C5.

Maximum output current is set with switches S1, S2 and S3 as shown in Table II.

Construction and testing

An actual-size, single-side PCB for the programmable power supply is shown in Fig. 2 and its component layout in Fig. 3. After assembling all components, enclose the PCB in a box such that 230V AC mains can be connected to the circuit easily. CON2, CON3 and CON4 can be mounted on the rear side of the cabinet. J1 through J3 shown in the PCB are simply jumper wires. Fix VR1 through VR6 on the front panel for setting output voltages.

Verify the test points given in Table III before using the circuit. After completing the circuit, connect CON5 to the MCU or printer port of the PC for setting various output voltages.